Finley Moss

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Education

University of Wisconsin–Madison College of Engineering — Expected May 2028
Bachelor of Science in Aerospace Engineering — Madison, WI

Engineering Experience

Undergraduate Student Assistant – Flight Simulation — July 2025 – Present — 10–30 hrs/week
UW–Madison College of Engineering — Madison, WI

- Designed, fabricated, assembled, and tested **three full-scale Cessna 172 G1000 flight simulators** for AE 423: Flight Dynamics and Controls, integrating X-Plane, flight controls, avionics, motion platforms, displays, and instrumentation.
- Developed simulator components from CAD designs through HDF prototypes and final fabrication, iterating designs based on testing, fit, system constraints, and functional requirements.
- Developed flight-data acquisition and telemetry workflows to collect, export, and analyze pitch, angle of attack, airspeed, and other aircraft parameters for flight-dynamics experiments.
- Built and configured high-performance simulator PCs and integrated computer hardware, avionics, displays, and simulation software.
- Supported procurement and project management for a **$60,000 engineering project**, tracking equipment, vendors, deliveries, and approximately $58,000 in expenditures.

Teaching Assistant – AE 423 Flight Controls and Dynamics — Jan 2026 – May 2026 — 10 hrs/week
UW–Madison College of Engineering — Madison, WI

- Supported laboratory instruction for senior-level aerospace engineering students on flight dynamics, aircraft maneuvers, and flight simulation.
- Operated flight simulators to conduct dynamic-system tests and demonstrate aircraft response to commanded maneuvers.
- Guided students in collecting and analyzing numerical flight data to connect simulated aircraft behavior with flight-dynamics theory.
- Developed standardized testing procedures to improve safety, repeatability, and data quality during laboratory exercises.

Engineering Ambassador — May 2026 – Aug 2026 — 3 hrs/week
UW–Madison College of Engineering — Madison, WI

- Led College of Engineering tours for groups of 20–30 prospective students and families, communicating technical and academic information to diverse audiences.
- Presented engineering and academic information to audiences of **70–120 people**, adapting technical explanations to different levels of knowledge and responding to questions.

Engineering Projects

Flow Visualization Wind Tunnel Project – 1st Place — Oct 2024 – Apr 2025
EMA 200 – UW–Madison College of Engineering — Madison, WI

- Designed, fabricated, and tested a wind tunnel to experimentally investigate aerodynamic behavior using flow-visualization techniques.
- Developed and executed an experimental testing approach, collected aerodynamic observations, and analyzed flow behavior to evaluate test results.
- Collaborated with a team to troubleshoot the test setup, interpret experimental results, and communicate findings at the UW–Madison Engineering Expo.
- Awarded **1st Place, Group 1** by the Engineering Expo Executives.

Leadership and Involvement

Theta Tau Xi Chapter – Treasurer — May 2026 – Present
Professional Engineering Fraternity — Madison, WI

- Manage chapter financial records, budgeting, reimbursements, payments, and financial planning for professional development, service, fundraising, and social activities.
- Coordinated chapter-wide fundraising operations that generated more than **$10,000** over five months, managing volunteers, events, vendors, and fundraising logistics.

American Institute of Aeronautics and Astronautics (AIAA) — Oct 2026 – Present
Fundraising Chair — Madison, WI

Technical Skills

Software: MATLAB, SolidWorks, X-Plane, Python, Microsoft Office, LaTeX
Fabrication: 3D Printing, Laser Cutting, Woodworking, Table Saw, Drill Press, Band Saw, Router, Power Tools